



Vascular Access & IV Fluid Therapy

NOTE: Do not access any existing port without direct online medical control approval (such as picc lines, central lines, fistulas, etc).

Indications

1. Patients with potential need for either fluid resuscitation or medication administration.
2. External jugular cannulation should be initiated in patients in whom access is necessary and other peripheral vascular access is not accessible or is contraindicated.
3. IO indications: Adult and pediatric life-threatening situations where venous access using peripheral veins has been unsuccessful. IO access should be considered early in situations where IV access is unsuccessful or technically challenging. Indications include:
 - A. Cardiac Arrest
 - B. Severe burn injury with shock
 - C. Shock
 - D. Severe multi-system trauma with shock
 - E. For other situations contact Medical Control. Do not delay transport.



Contraindications

1. To peripheral vascular access:
 - A. No peripheral sites available
 - B. Burns overlying available peripheral sites unless no other sites available
 - C. Infection overlying available peripheral sites
2. To intraosseous infusion and placement:
 - A. Infiltration of previously placed IO. If infiltration occurs (rare), do not reuse the same bone as fluid will leak out of the original hole; select another site.
 - B. Placement in fractured extremity. If the femur is fractured do not use the tibia of same leg.
 - C. Burns overlying available peripheral sites unless no other sites available
 - D. Infection overlying available peripheral sites
3. To fluid bolus:
 - A. Pulmonary edema
 - a. Contact Medical Control when pulmonary edema is present, yet clinical presentation indicates the need for fluid resuscitation.



Special Considerations (Side effects/Complications)

1. Initiation of vascular access generally should not delay patient transport to the hospital.
2. General side effects or complications: infection, air embolism, catheter shear, hematoma, arterial puncture, and fluid overload.
3. Intraosseous placement:

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- A. Complications include subperiosteal infusion, osteomyelitis, sepsis, fat embolism, and bone marrow damage.

Standards for IV attempts

1. Two (2) attempts per provider, maximum 4 attempts.
2. Consider IO early, as indicated above.
3. Document any reasons for deviation.

Needle size for IV placement

1. Adult 18 ga - 20 ga angiocath
2. Adult uncompensated shock or cardiac arrest 14 ga - 18 ga angiocath.
3. Pediatrics 20 ga - 24 ga angiocath

Solutions – Unless otherwise specified, the IV solution may be **normal saline 0.9% (NS)** or **lactated ringers (LR)**. **NS** is to be used for dilution and/or reconstitution unless otherwise specified in applicable protocol.

Flow Rates and Volume

1. Saline lock is preferred, unless fluid administration is needed.
2. Flow rates, changes in flow rates, and total volume administered must be documented on the EMS Patient Care Record.
3. Fluid Bolus – for fluid resuscitation (i.e., dehydration, hypotension, etc.)
 - a. Adults (>14 years of age): 1 liter IV/IO wide open with repeat of 1 additional liter as necessary (maximum total of 2L), unless otherwise noted by protocol.
 -  b. Pediatrics (≤ 14 years of age): 20 mL/kg IV/IO wide open with repeat of 20 mL/kg as necessary (maximum total of 40 mL/kg), unless otherwise noted by protocol,
4. IV/IO fluid bolus is contraindicated in patients with pulmonary edema.
5. Medicated drips should be piggybacked into a **NS** main IV line or saline lock.
6. Medications should be administered via a saline lock (preferred).
7. If the main IV running is **LR**:
 - a. Prior to administration of any medication, the **LR** IV will be paused.
 - b. The line will be flushed with 5-10 ml of saline flush.
 - c. The medication will be administered.
 - d. The line will be flushed with 5-10 ml of saline flush.
 - e. The **LR** IV will be resumed.

IV Tubing

1. Macro drip is the preferred tubing.

Procedure IV/IO Placement

1. Utilize universal precautions for all IV/IO placements.

Procedure for Peripheral Vascular Cannulation:

1. Gather and prepare equipment.

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2. Place the tourniquet on the extremity.
3. Cleanse the skin
4. Make your puncture while maintaining vein stability.
5. Watch for flashback. Once you have a blood return, advance the catheter as per normal IV technique and attach the IV tubing or saline lock tubing and cap.
6. If you have no blood return and you are in the vein, remove the needle hub and attach your syringe to assist in aspirating for blood.
7. Instill 2-3 mL of normal saline if normal saline lock placed.
8. Secure catheter and IV tubing.

Procedure for External Jugular Cannulation:

1. Gather and prepare equipment
2. Position patient supine (Trendelenburg, if possible)
3. Turn head to opposite side of venipuncture (if no C-spine injury is suspected)
4. Cleanse the skin
5. Occlude the vein by using the side of your finger above the clavicle to facilitate filling the vein.
6. Make your puncture midway between the angle of the jaw and the middle of the clavicle.
7. Watch for flashback. Once you have a blood return, advance the catheter as per normal IV technique and attach the IV solution or normal saline lock cap, covering catheter with gloved finger while preparing to attach the IV tubing. If you have no blood return and you are in the vein, remove the needle hub and attach your syringe to assist in aspirating for blood.
8. Instill 2-3 mL of normal saline if normal saline lock placed.
9. Secure IV catheter and tubing.

Procedure for Intraosseous Placement:

1. Have all IO equipment ready prior to bone penetration.
2. Expose the extremity.
3. Stabilize the extremity to minimize motion.
4. Selection of site:
 - A. Medial aspect of proximal tibia or proximal humerus.
 - B. In children less than six years of age, the preferred site is the proximal tibia.
 - C. In cardiac arrest, the preferred site is the proximal humerus.
5. Insertion:
 - A. Follow the manufacturer's recommendations for IO insertion with the above indications.
6. Scrub the insertion site with alcohol prep/chlorhexidine. Strict adherence to aseptic technique is essential.
7. Insert the IO needle.
8. Attempt to confirm marrow placement by removing the stylet and aspirating blood and/or bone marrow.
 - A. If unable to aspirate, attach 10 – 20 mL syringe with **NS** and gently infuse fluid.

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- B. Observe for normal saline leakage or SQ tissue swelling.
 - a. If neither occurs, proceed.
 - b. If occurs, select a different site.
- 9. Connect the appropriate IV equipment (normal saline locks not indicated in IO placement).
-  10. In conscious patients, prior to initiating infusion, may administer **lidocaine 2%**
 - a. Adult 20 mg IO
 -  b. Pediatrics 0.5 mg/kg, IO maximum dose of 20 mg.
- 11. Administer the appropriate fluids and/or drugs.
- 12. Stabilize the entire intraosseous set-up as if securing an impaled object.
-  13. Conscious patients experiencing pain with infusion may require an additional dose of **lidocaine 2%**.
-  14. If the IO is unsuccessful after 2 attempts, contact Medical Control

Procedure for Discontinuation/Nonfunctioning IVs:

- 1. Patients who have refused transport that have received an IV must have the IV removed and it must be documented in the patient care record.
- 2. Nonfunctioning IVs will be handled in one of the following ways:
 - a. Removed prior to transfer of care - placement and removal of the nonfunctioning IV will be included in both verbal and written reports.
 - b. Not removed prior to transfer of care – the nonfunctioning IV will be clearly marked as nonfunctioning and it will be included in both verbal and written reports.

Medication Protocols

Lidocaine

Reference(s):

Vallée [et.al.](#) Compatibility of Lactated Ringer's Injection With 94 Selected Intravenous Drugs During Simulated Y-site Administration. Hospital Pharmacy 2021, Vol. 56(4) 228–234.